Defend and revise a forecast

LEARNING TARGET I can defend a prediction, state limits and revise it.

A · KEY VOCABULARY

Prediction	Result, units, time, direction and range.
Evidence	A measurement or calculation supporting a step.
Limit	An uncertain input or missing effect.
Revision	Name the change and show its effect.

B · KEY EQUATIONS + UNITS

$$x_f = x_0 + vt$$

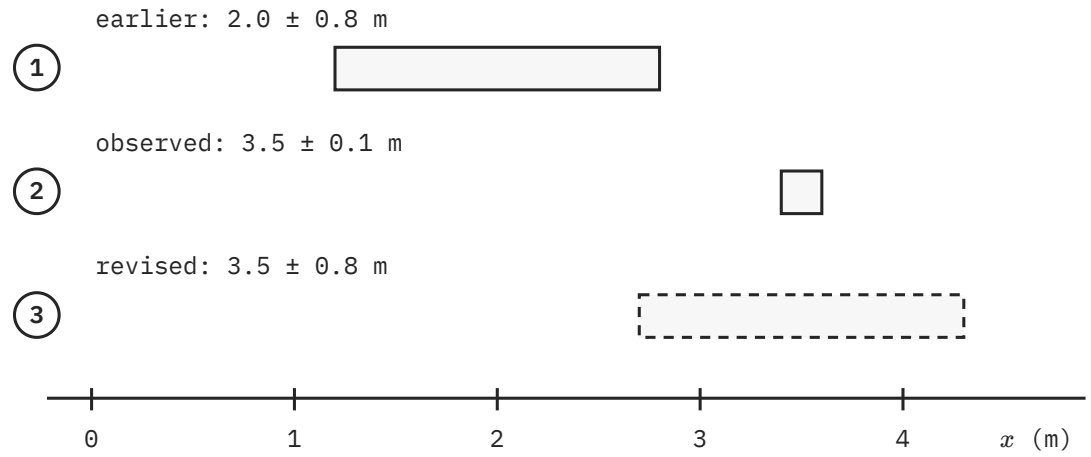
$$\delta x_f = \delta x_0 + t\delta v$$

t exact time

bounds not probabilities

C · COLOR IT · COMPARE THE RANGES

Illustrative data; same axis.



1 · BLUE Earlier forecast

2 · BLUE Observed range

3 · BLUE Revised forecast

D · WORKED EXAMPLE

Keep x_0 = **8 ± 0.5 m** and exact t = **3 s**. A new measurement changes v from −2.0 to **−1.5 ± 0.1 m/s**. **Update the forecast.**

GIVEN	EQUATION	WORK	ANSWER
x_0 = **8 ± 0.5 m** v = **−1.5 ± 0.1 m/s** exact t = **3 s**	Update center and conservative bound.	x_f = 8 − 1.5(3) δx = .5 + 3(.1) = **0.8 m**	**2.7 to 4.3 m** Center shifts +**1.5 m**.

E · THE 2026-XJ THREAD

DAY 22 OF 22 · NEW TOOL: A REVISED FORECAST

Show the evidence behind XJ's forecast and what changed after review. Separate the physics from cost and feasibility decisions.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else